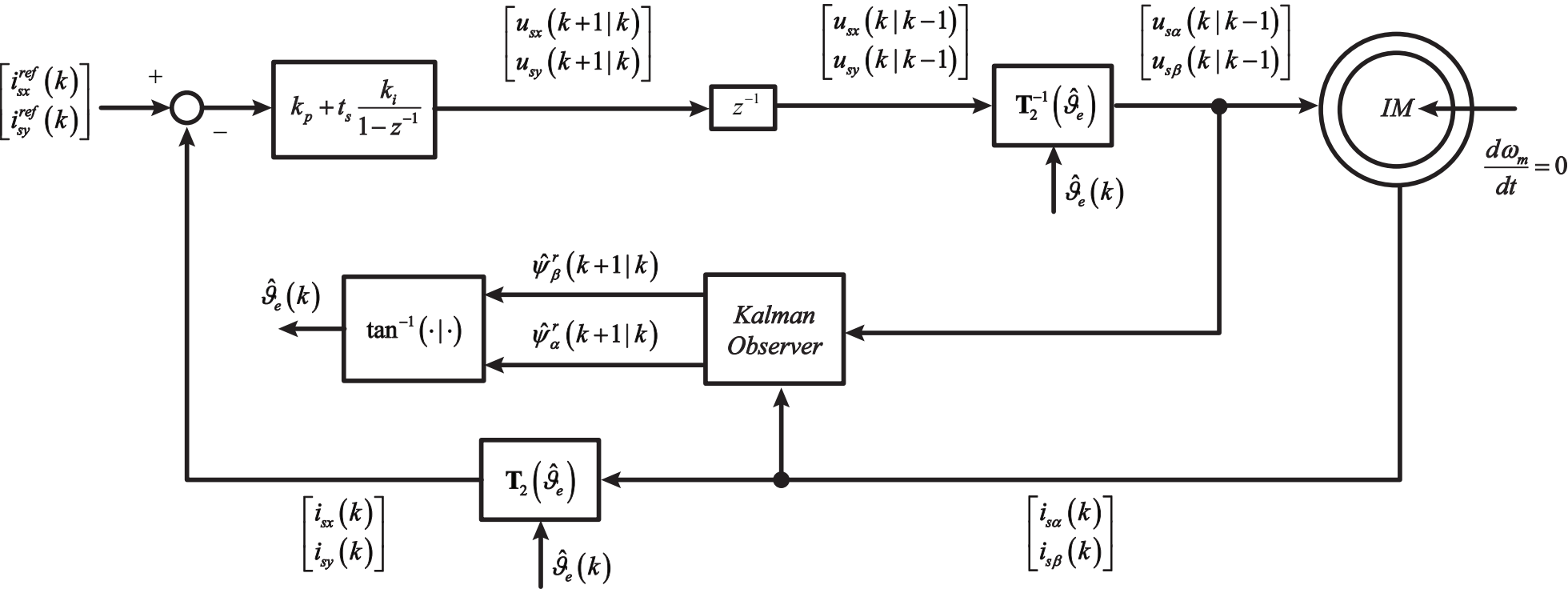
$\begin{bmatrix} i_{sx}^{ref}(k) \\ i_{sy}^{ref}(k) \end{bmatrix}$
+
−
$k_p + t_s \frac{k_i}{1-z^{-1}}$
$\begin{bmatrix} u_{sx}(k+1|k) \\ u_{sy}(k+1|k) \end{bmatrix}$
z^{-1}
$\begin{bmatrix} u_{sx}(k|k-1) \\ u_{sy}(k|k-1) \end{bmatrix}$
$\mathbf{T}_2^{-1}(\hat{\vartheta}_e)$
$\hat{\vartheta}_e(k)$
$\begin{bmatrix} u_{s\alpha}(k|k-1) \\ u_{s\beta}(k|k-1) \end{bmatrix}$
IM
$\frac{d\omega_m}{dt} = 0$
$\hat{\psi}_\beta^r(k+1|k)$
$\hat{\psi}_\alpha^r(k+1|k)$
Kalman Observer
$\tan^{-1}(\cdot|\cdot)$
$\hat{\vartheta}_e(k)$
$\mathbf{T}_2(\hat{\vartheta}_e)$
$\hat{\vartheta}_e(k)$
$\begin{bmatrix} i_{sx}(k) \\ i_{sy}(k) \end{bmatrix}$
$\begin{bmatrix} i_{s\alpha}(k) \\ i_{s\beta}(k) \end{bmatrix}$